PhyloTrace

APPLICATION NOTE

Retrospective analysis of a decade-long *Pseudomonas aeruginosa* outbreak

Summary. PhyloTrace typed a published collection of 253 outbreak genomes on a single desktop workstation in 49 minutes at default settings, resolved the outbreak as one cluster at the scheme's published threshold and detected the two chromosomal resistance determinants behind its resistance evolution, with the subclone distribution reported in the original study.

253	3,867	ST-621	4,176
genomes	cgMLST loci per genome	classical MLST, 252 of 253 isolates	resistance genes and mutations detected

Background

Healthcare-associated *P. aeruginosa* outbreaks can persist for years, often fed by environmental reservoirs such as sink drains. Genomic surveillance can reveal these links — but only if laboratories can analyse genomes together with their epidemiological context.

A published, independently analysed collection documents such an outbreak in a 425-bed US hospital, spread across 26 wards over ten years and sustained by reservoirs in sink drains.[1] We used it to test whether PhyloTrace reproduces the published findings under routine conditions.

Data and approach

Dataset	253 genome assemblies (ST-621) from 82 patients, collected May 2011 – December 2020 across 26 wards of a 425-bed US hospital · NCBI BioProject PRJNA852179[1]
Typing	cgMLST (3,867 loci, *P. aeruginosa* scheme, version 1.0 from cgMLST.org, published cluster threshold 12) and classical MLST
Screening	NCBI AMRFinderPlus for resistance genes, virulence factors and point mutations
Settings	Default settings, no manual tuning
Metadata	Collection date, specimen source, patient, building, floor and ward from the published study
Hardware	Desktop workstation (AMD Ryzen 7 5800X, 8 cores, 24 GB RAM, Linux) · PhyloTrace 2.0.0

Results

Fast, unattended typing. All 253 genomes were typed and screened in 49 minutes on a desktop workstation — a median of 12 seconds per genome, at a median locus completeness of 99.6 %.

Sequence type as published. Classical MLST assigned ST-621 to 252 of the 253 isolates, each an exact match to a registered PubMLST profile — the sequence type reported in the original study.

One outbreak cluster. At the scheme's published threshold, the collection resolved into a single outbreak cluster.

Resistance determinants recovered. The two chromosomal changes behind the outbreak's resistance evolution were detected: the *oprD* stop codon *oprD*_W6STOP in 114 isolates (113 of them subclone SC2) and the PBP3 substitution *ftsI*_R504C in 87 isolates (79 SC1, 8 SC2). Their distribution matches the original study. As there, no acquired carbapenemase or extended-spectrum β-lactamase gene was found.

Genotype and phenotype side by side. With the published susceptibility results entered as variables, determinant and phenotype were compared isolate by isolate: for ciprofloxacin they agreed in 99.6 % of the 250 screened isolates. For imipenem, agreement was lower, because the *oprD* frameshift reported in subclone SC1 is outside what the screen detects — a limit that shows up directly in the figures.

Epidemiological context. Collection date, ward, building and specimen source were added as variables to the typing results. Epidemic curves stratified by subclone and by building (Figures 5 and 6) show how the outbreak developed over the decade. All six figures follow on pages 3 and 4.

What this means for your laboratory

One application, not four tools

The original study combined four separately operated tools and reconciled them with the metadata by hand.[1] PhyloTrace covered the same scope in one.

In-house, without writing code

From assemblies to clusters, resistance and context on one workstation — no upload to external servers.

Ready for ongoing monitoring

Saved analyses reopen as new isolates arrive; every result keeps its scheme version, thresholds and tool versions.

References

[1] Stribling, W. et al. (2025) Detecting, mapping, and suppressing the spread of a decade-long *Pseudomonas aeruginosa* nosocomial outbreak with genomics. *eLife* 13, RP93181. doi:10.7554/eLife.93181. Data: NCBI BioProject PRJNA852179.
[2] PhyloTrace manuscript in preparation.

Request a demo
info@liora-bioinformatics.com · liora-bioinformatics.com/phylotrace

For research use only. Not a medical device; not intended for diagnostic procedures or patient management.

FIGURES 1–4

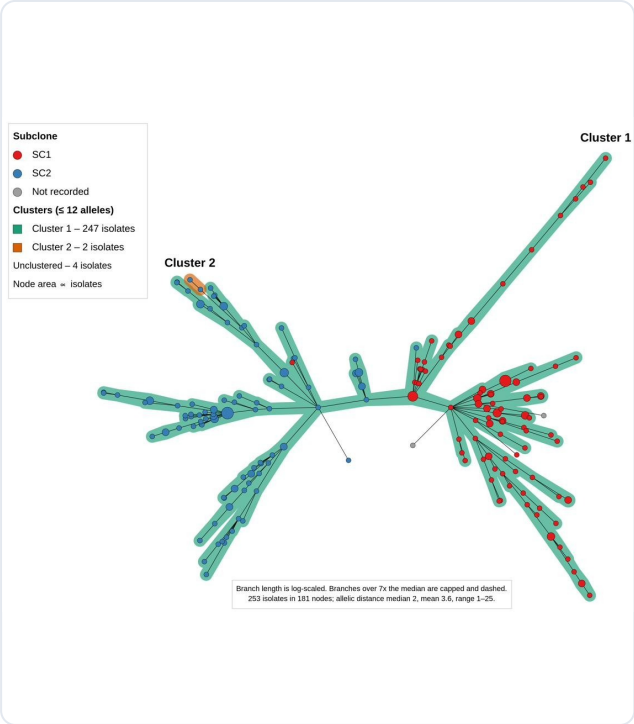


Figure 1. Minimum spanning tree of the 253 isolates. Node colour shows the published subclone; clusters are shaded at the scheme's published threshold of 12 alleles; node area is proportional to the number of isolates a node holds.

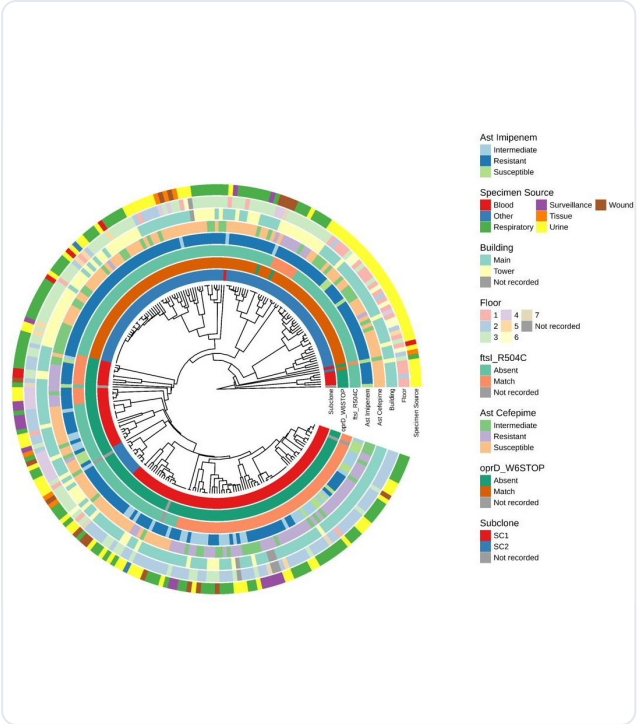


Figure 2. Neighbour-Joining tree, circular layout. Annotation rings from outside to inside: specimen source, floor, building, cefepime and imipenem phenotype, *ftsI*_R504C, *oprD*_W6STOP, subclone.

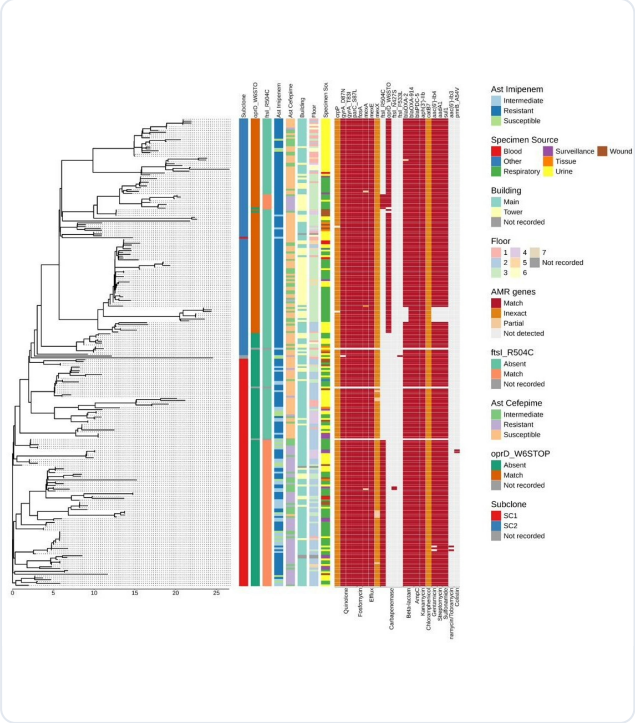


Figure 3. Neighbour-Joining tree, rectangular layout, with tip-aligned annotations and a panel of every detected resistance element grouped by drug class.

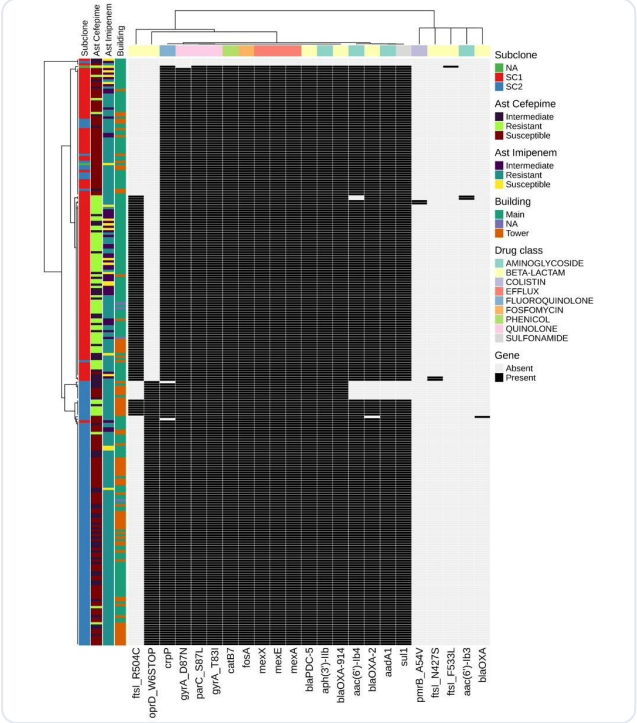


Figure 4. Resistance heatmap of the 250 screened isolates. Presence and absence of each element, columns grouped by drug class, rows annotated with subclone, building and phenotype.

FIGURES 5–6

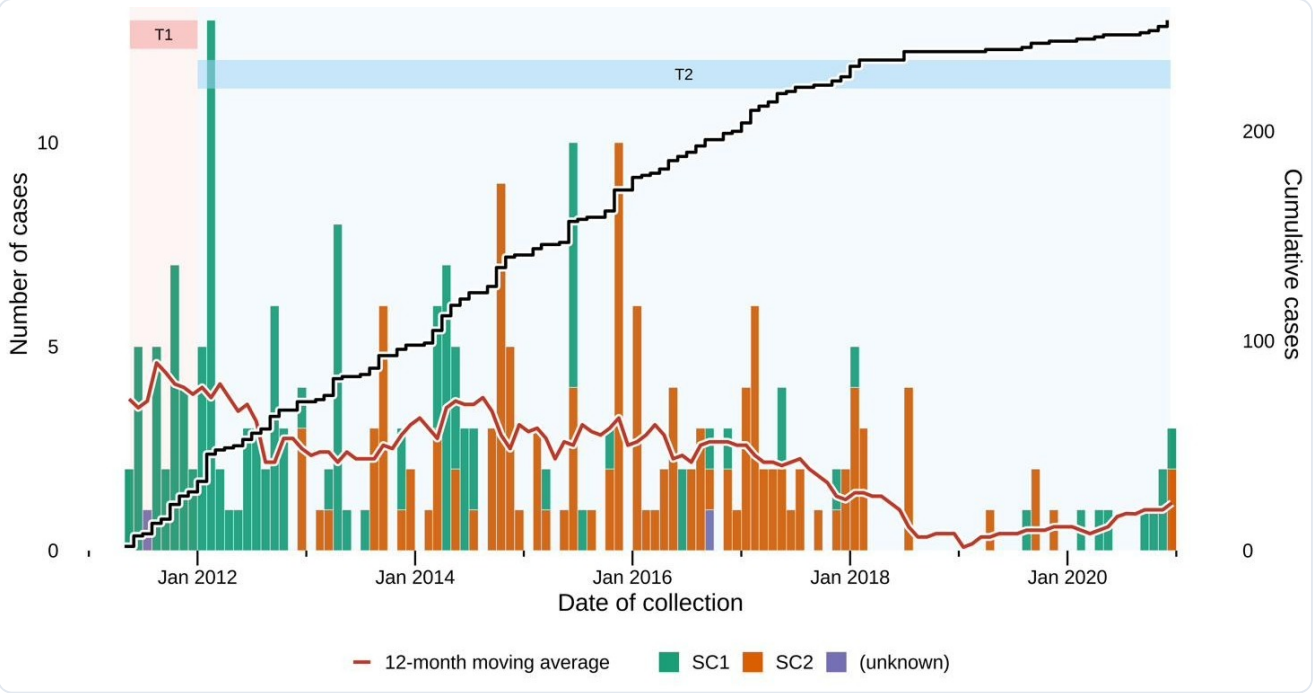


Figure 5. Epidemic curve 2011–2020, binned by month and stratified by subclone, with a 12-month moving average and cumulative cases. T1 and T2 mark the opening of the Main Building and the Tower Building extension in 2012.

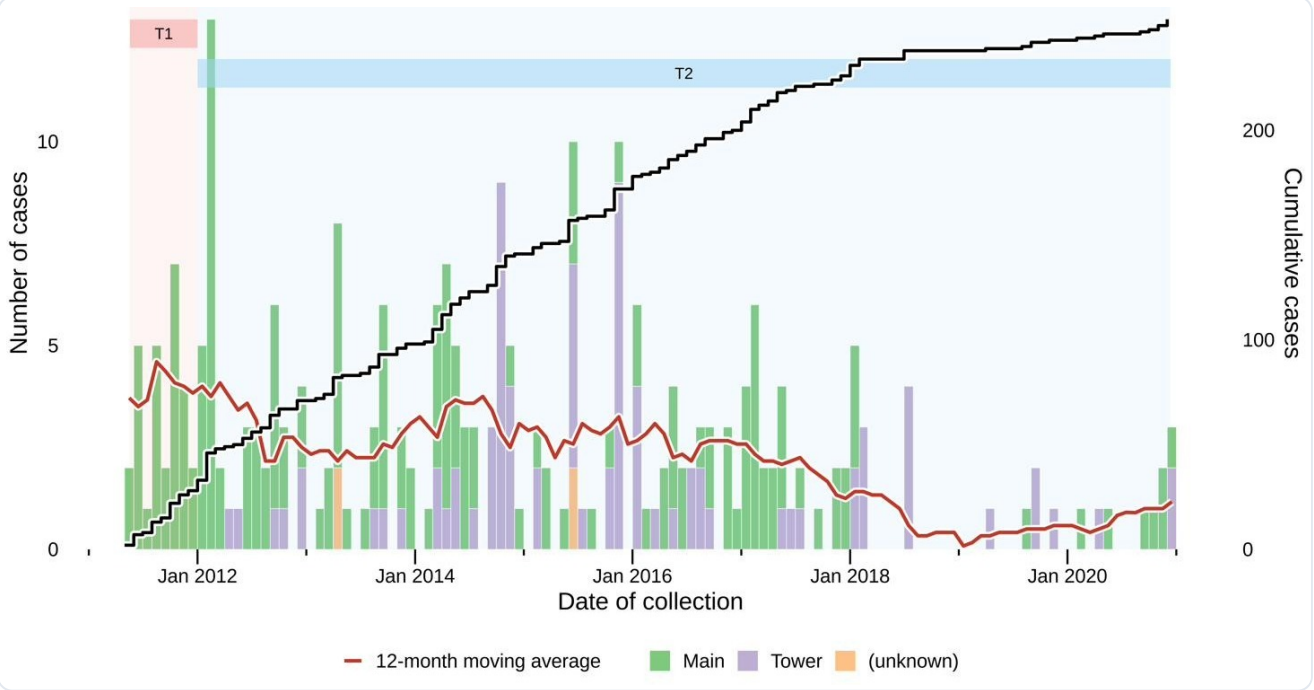


Figure 6. Epidemic curve stratified by building. The first Tower Building isolate was collected on 3 April 2012.